



**Generation Interconnection
System Impact Study Report
for
Queue Project AF2-143
POWERTON-NEVADA 345 KV
90 MW Capacity / 150 MW Energy**

February 2021

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is ComEd.

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 General

The Interconnection Customer (IC), has proposed a Solar generating facility located in Lasalle County, Illinois. The installed facilities will have a total capability of 150 MW with 90 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this project is December 31, 2023. This study does not imply a TO commitment to this in-service date.

Queue Number	AF2-143
Project Name	POWERTON-NEVADA 345 KV
State	Illinois
County	Lasalle
Transmission Owner	ComEd
MFO	150
MWE	150
MWC	90
Fuel	Solar
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

AF2-143 will interconnect with the ComEd on transmission system at the AD2-038 substation which taps the Powerton - Goodings Grove 345 kV line.

5 Cost Summary

The AF2-143 project will be responsible for the following costs:

Description	Cost
Total Physical Interconnection Costs	\$ 200,000
Allocation towards System Network Upgrade Costs*	\$ 16,573,400
Total Costs	\$ 16,773,400

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

6.1 Attachment Facilities

To accommodate interconnection of AF2-143; the relaying, SCADA, communication, and metering between Osagrove Flats Wind Farm and TSS 920 Dana would be reviewed and upgraded if needed.

ComEd would take approximately 18-months to review and possibly upgrade the relaying, SCADA, Communication and metering after the ISA / ICSA are signed. See Section 11.5 for System Upgrades schedule.

The total physical interconnection costs is given in the table below:

Description	Cost
The relaying, SCADA, communication, and metering between Osagrove Flats Wind Farm and TSS 920 Dana would be reviewed and upgraded if needed.	\$200,000
Total Physical Interconnection Costs	\$200,000

7 Schedule

See Sections 6 and 11.

8 Transmission Owner Analysis

See Section 6.

9 Interconnection Customer Requirements

The Interconnection Customer is responsible for all design and construction related activities on the Interconnection Customer's side of the Point of Interconnection.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Meteorological Data Reporting Requirements

The solar generation facility shall provide the Transmission Provider with site-specific meteorological data including:

- Temperature (degrees Fahrenheit)
- Atmospheric pressure (hectopascals)
- Irradiance
- Forced outage data

10.3 Interconnected Transmission Owner Requirements

The IC will be required to comply with all Interconnected Transmission Owner's revenue metering requirements for generation interconnection customers located at the following link:

ComEd interconnection requirements can be found at <https://www.pjm.com/planning/design-engineering/to-tech-standards/private-comed.aspx>

To the extent that these Applicable Technical Requirements and Standards may conflict with the terms and conditions of the Tariff, the Tariff shall control.

11 Summer Peak Analysis

The Queue Project AF2-143 was evaluated as a 150.0 MW (Capacity 90.0 MW) injection at the AD2-038 substation which taps the Powerton - Goodings Grove 345 kV line in the ComEd area. Project AF2-143 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF2-143 was studied with a commercial probability of 100%. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE CT LOADIN G %	POST PROJE CT LOADIN G %	AC DC	MW IMPACT
9626422	270716	DRESDE N ; B	345.0	CE	270928	WOLFS ; B	345.0	CE	1	COMED_P4_012-45-BT5-6	breaker	1479.0	99.23	100.07	AC	11.62
96265467	918050	AA1-018 TAP	345.0	CE	270769	GOODING S ; 2R	345.0	CE	1	COMED_P7_345-L1202__B-S_+345-L1227__R-S-A	tower	1494.0	97.34	101.66	AC	62.41

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE CT LOADIN G %	POST PROJE CT LOADIN G %	AC DC	MW IMPACT
96263720	270770	GOODIN GS ; 4B	345.0	CE	270766	GOODIN GS ; 3B	345.0	CE	1	COMED_P2-2_116_GG-345R_2	bus	1802.0	113.81	114.49	AC	21.58
96264009	270770	GOODIN GS ; 4B	345.0	CE	270766	GOODIN GS ; 3B	345.0	CE	1	COMED_P4_116-45-L11614_	breaker	1802.0	118.53	119.21	AC	21.56
96264010	270770	GOODIN GS ; 4B	345.0	CE	270766	GOODIN GS ; 3B	345.0	CE	1	COMED_P4_116-45-L9801__FSA	breaker	1802.0	113.81	114.49	AC	21.58
96264011	270770	GOODIN GS ; 4B	345.0	CE	270766	GOODIN GS ; 3B	345.0	CE	1	COMED_P4_116-45-TR82__	breaker	1802.0	113.81	114.49	AC	21.58
96263945	270855	POWER TON ; R	345.0	CE	349662	7TAZE W ELL	345.0	AMI L	1	COMED_P4_012-45-BT4-5__	breaker	1489.0	121.64	126.39	AC	69.87
96265327	270855	POWER TON ; R	345.0	CE	349662	7TAZE W ELL	345.0	AMI L	1	COMED_P7_345-L19601_B-S_+345-L9801__R-S_FSA	tower	1489.0	135.29	142.55	AC	110.8
148526155	270855	POWER TON ; R	345.0	CE	349662	7TAZE W ELL	345.0	AMI L	1	COMED_P7_345-L0301__B-S_+345-L9801__R-S_FSA	tower	1489.0	120.74	127.94	AC	110.83
96265342	930740	AB1-122 TAP	345.0	CE	270717	DRESDE N ; R	345.0	CE	1	COMED_P7_345-L19601_B-S_+345-L9801__R-S_FSA	tower	1195.0	132.72	135.16	AC	32.08
148526175	930740	AB1-122 TAP	345.0	CE	270717	DRESDE N ; R	345.0	CE	1	COMED_P7_345-L0301__B-S_+345-L9801__R-S_FSA	tower	1195.0	127.76	130.23	AC	32.09
96265405	930750	AB1-122 TAP	345.0	CE	270716	DRESDE N ; B	345.0	CE	1	COMED_P7_345-L19601_B-S_+345-L9801__R-S_FSA	tower	1768.0	106.08	108.01	AC	37.91
148526276	930750	AB1-122 TAP	345.0	CE	270716	DRESDE N ; B	345.0	CE	1	COMED_P7_345-L0301__B-S_+345-L9801__R-S_FSA	tower	1768.0	102.1	104.03	AC	37.92

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE CT LOADI NG %	POST PROJE CT LOADI NG %	AC DC	MW IMPA CT
95294071	960680	AF2-359 TAP	345.0	CE	243229	05OLIVE	345.0	AEP	1	AEP_P4_#2978_05DUMONT_NON_FSA	breaker	971.0	165.71	165.73	AC	15.37
95294072	960680	AF2-359 TAP	345.0	CE	243229	05OLIVE	345.0	AEP	1	AEP_P4_#2978_05DUMONT_FSA	breaker	971.0	165.73	165.75	AC	15.37

11.4 Steady-State Voltage Requirements

None

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE CT LOADI NG %	POST PROJE CT LOADI NG %	AC DC	MW IMPA CT
96264466	270855	POWERTON ; R	345.0	CE	349662	7TAZEWE LL	345.0	AMILL	1	COMED_P1-2_345-L1202__B-S-A	operation	1489.0	121.05	125.83	AC	69.99
96264416	274750	CRETE EC ; BP	345.0	CE	255112	17STJOHN	345.0	NIPS	1	Base Case	operation	1091.0	142.05	142.18	AC	12.0
96264548	930740	AB1-122 TAP	345.0	CE	270717	DRESDEN ; R	345.0	CE	1	COMED_P1-2_345-L1202__B-S-A	operation	1195.0	121.9	123.46	AC	18.67
96264542	930750	AB1-122 TAP	345.0	CE	270716	DRESDEN ; B	345.0	CE	1	COMED_P1-2_345-L0304__R-S	operation	1479.0	115.36	118.08	AC	40.19
96264543	930750	AB1-122 TAP	345.0	CE	270716	DRESDEN ; B	345.0	CE	1	Base Case	operation	1201.0	115.97	117.45	AC	17.76

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-143	Upgrade Number																																				
96264242	1	DRESDEN ; B 345.0 kV - WOLFS ; B 345.0 kV Ckt 1	The applicable rating to apply is the SSTE. ComEd 345kV L1221 SSTE rating is 1568 MVA. No upgrade required.	\$0	\$0	N/A																																				
96265405,1485 26276	6	AB1-122 TAP 345.0 kV - DRESDEN ; B 345.0 kV Ckt 1	The applicable rating to apply is the ALDR. ComEd 345kV L1202 ALDR rating is 2033 MVA. No upgrade required.	\$0	\$0	N/A																																				
148526175,962 65342	5	AB1-122 TAP 345.0 kV - DRESDEN ; R 345.0 kV Ckt 1	The upgrade will be to reconductor the line. The estimate for this upgrade is \$6.925 M with a preliminary construction timeline of 30-36 months. Upon completion of this upgrade the new ratings will be 1334/1726/1837/2084 MVA (SN/SLTE/SSTE/SLD). PJM Network Upgrade N5756. The cost allocation is: <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>\$ cost (\$6.925 M)</th></tr><tr><td>AB1-122</td><td>560.0</td><td>64.65%</td><td>4.4773</td></tr><tr><td>AD1-039</td><td>51.5</td><td>5.94%</td><td>0.4113</td></tr><tr><td>AE1-113</td><td>36.4</td><td>4.20%</td><td>0.2910</td></tr><tr><td>AE1-163</td><td>78.1</td><td>9.02%</td><td>0.6244</td></tr><tr><td>J974</td><td>47.3</td><td>5.46%</td><td>0.3782</td></tr><tr><td>J1055</td><td>17.1</td><td>1.97%</td><td>0.1367</td></tr><tr><td>AF2-142</td><td>37.9</td><td>4.38%</td><td>0.3030</td></tr><tr><td>AF2-143</td><td>37.9</td><td>4.38%</td><td>0.3031</td></tr></table>	Queue	MW contribution	Percentage of Cost	\$ cost (\$6.925 M)	AB1-122	560.0	64.65%	4.4773	AD1-039	51.5	5.94%	0.4113	AE1-113	36.4	4.20%	0.2910	AE1-163	78.1	9.02%	0.6244	J974	47.3	5.46%	0.3782	J1055	17.1	1.97%	0.1367	AF2-142	37.9	4.38%	0.3030	AF2-143	37.9	4.38%	0.3031	\$6.925 M	\$303.1 K	N5756
Queue	MW contribution	Percentage of Cost	\$ cost (\$6.925 M)																																							
AB1-122	560.0	64.65%	4.4773																																							
AD1-039	51.5	5.94%	0.4113																																							
AE1-113	36.4	4.20%	0.2910																																							
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J974	47.3	5.46%	0.3782																																							
J1055	17.1	1.97%	0.1367																																							
AF2-142	37.9	4.38%	0.3030																																							
AF2-143	37.9	4.38%	0.3031																																							

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-143	Upgrade Number																								
96263945,1485 26155,9626532 7	4	POWERTON ; R 345.0 kV - 7TAZEWELL 345.0 kV Ckt 1	ComEd end: ComEd-end ALDR is 1838 MVA. Upgrade will be to mitigate sag on ComEd portion of line. A preliminary estimate is \$9.1 M with an estimated construction timeline of 24 months. Upon completion the ComEd ratings are expected to be 1461/1656/1909/1912/2198 MVA (SN/SLTE/SSTE/SLD/ALDR). N6280. The cost allocation is: <table><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>\$ cost (\$ 9.1 M)</th></tr><tr><td>AE1-163</td><td>46.6</td><td>13.37%</td><td>1.2165</td></tr><tr><td>AE2-255</td><td>44.12</td><td>12.65%</td><td>1.1509</td></tr><tr><td>AE2-281</td><td>36.5</td><td>10.46%</td><td>0.9521</td></tr><tr><td>AF2-142</td><td>110.8</td><td>31.76%</td><td>2.8903</td></tr><tr><td>AF2-143</td><td>110.8</td><td>31.76%</td><td>2.8903</td></tr></table> MISO/Ameren-end: Ameren limit is 1512 MVA due to 2 conductor 12777 ACAR. Would need to reconductor or rebuild 0.64 miles of conductor with 2x954 Kcmil ACSR 45/7 RAIL and replace bus conductor to 3500 A. \$750K. Time estimate: 24 months. New SE rating 2091 MVA. This upgrade is driven by a prior queue cycle. Note: an additional Ameren upgrade is required to to cover the flow of 2123 MVA. This will be evaluated in the Facilities Study along with other MISO impacts.	Queue	MW contribution	Percentage of Cost	\$ cost (\$ 9.1 M)	AE1-163	46.6	13.37%	1.2165	AE2-255	44.12	12.65%	1.1509	AE2-281	36.5	10.46%	0.9521	AF2-142	110.8	31.76%	2.8903	AF2-143	110.8	31.76%	2.8903	\$9.1 M \$750 K	\$2.8903 M TBD	N6280
Queue	MW contribution	Percentage of Cost	\$ cost (\$ 9.1 M)																											
AE1-163	46.6	13.37%	1.2165																											
AE2-255	44.12	12.65%	1.1509																											
AE2-281	36.5	10.46%	0.9521																											
AF2-142	110.8	31.76%	2.8903																											
AF2-143	110.8	31.76%	2.8903																											
96264010,9626 4011,96264009, 96263720	3	GOODINGS ;4B 345.0 kV - GOODINGS ;3B 345.0 kV Ckt 1	The applicable rating to apply is the SSTE. ComEd TSS 116 345 kV BT 3-4 SSTE rating is 2083 MVA. The upgrade will be a 345kV BT 3-4 circuit breaker replacement and station conductor. A preliminary estimate for this upgrade is \$3.2 M with a preliminary construction timeline of 24 months. Upon completion of the upgrade the new ratings are expected to be 1793/2008/2425/2918 MVA (SN/SLTE/SSTE/SLD). This upgrade is presently driven by a prior queue cycle.	\$3.2 M	\$0	N6279																								

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-143	Upgrade Number																				
95294071,95294072	7	AF2-359 TAP 345.0 kV - 05OLIVE 345.0 kV Ckt 1	AEP: SE rating is 971 MVA. A sag check will be required for the ACSR/PE ~ 1414 ~ 62/19 ~ Conductor Section 1 to determine if the line section can be operated above its emergency rating of 971 MVA. The results could prove that no additional upgrades are necessary, that some upgrades on the circuit are necessary, or that the entire 40.61 mile section of line would need to be rebuilt. Estimated Cost: \$162,440. If deemed necessary to rebuild the entire 40.61 miles of the section of the line, Estimated Cost: \$81,220,000. Schedule: (1) Sag Study: 6 to 12 months. (2) Rebuild: The standard time required for construction differs from state to state. An approximate construction time would be 24 to 36 months after signing an interconnection agreement. New AEP end ratings to be 971/1304 MVA SN/SE. PJM Network Upgrade N4057. The N4057 upgrade is needed for a prior queue cycle. Replace 5 Jumpers (Sub cond 2156 ACSR 84/19 STD at Olive station, estimated cost: \$175,000. Perform an engineering study to determine if the Olive Relay Compliance Trip limits settings can be adjusted to mitigate the overload, Estimated Cost: \$25,000. New relay packages will be required if the settings cannot be adjusted, Estimated Cost: \$600,000. PJM Network Upgrade N5914. Rebuild/Reconductor 40.61 miles of the AEP owned section of the Olive - University Park (CE) 345 kV ACSR/PE 1414 62/19 line section 1. Estimated Cost to reconductor/rebuild AEP section of line: \$81.2 Million. The Olive switches to Line Riser will have to be replaced; Estimated Cost: \$1.4 Million. PJM Network Upgrade N5303 The cost allocation is: <table><thead><tr><th>Queue</th><th>MW contribution</th><th>Percentage of Cost</th><th>\$ cost (\$82.8 M)</th></tr></thead><tbody><tr><td>AF2-095</td><td>21.5</td><td>22.56%</td><td>18.680</td></tr><tr><td>AF2-096</td><td>41.4</td><td>43.44%</td><td>35.970</td></tr><tr><td>AF2-142</td><td>17.0</td><td>17.84%</td><td>14.770</td></tr><tr><td>AF2-143</td><td>15.4</td><td>16.16%</td><td>13.380</td></tr></tbody></table>	Queue	MW contribution	Percentage of Cost	\$ cost (\$82.8 M)	AF2-095	21.5	22.56%	18.680	AF2-096	41.4	43.44%	35.970	AF2-142	17.0	17.84%	14.770	AF2-143	15.4	16.16%	13.380	\$162,440 \$175 K \$25 K \$82.6 M	\$0 \$13.38 M	N4057 N5914 N5303
Queue	MW contribution	Percentage of Cost	\$ cost (\$82.8 M)																							
AF2-095	21.5	22.56%	18.680																							
AF2-096	41.4	43.44%	35.970																							
AF2-142	17.0	17.84%	14.770																							
AF2-143	15.4	16.16%	13.380																							

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF2-143	Upgrade Number
96265467	2	AA1-018 TAP 345.0 kV - GOODINGS ;2R 345.0 kV Ckt 1	The applicable rating to apply is the ALDR. ComEd 345kV L9806 ALDR rating is 1718 MVA. No upgrade required.	\$0	\$0	N/A
Total Cost				\$102,937,440	\$16,573,400	

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
96264242	270716	DRESDEN ; B	CE	270928	WOLFS ; B	CE	1	COMED_P4_012-45-BT5-6__	breaker	1479.0	99.23	100.07	AC	11.62

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274658	DRESDEN ;2U	510.5352	50/50	510.5352
274677	POWERTON ;5U	14.1420	50/50	14.1420
274678	POWERTON ;6U	14.0141	50/50	14.0141
274836	EQUISTAR ; R	4.7860	50/50	4.7860
274879	MINONK ;1U	2.0802	50/50	2.0802
276165	Z1-106 BAT2	0.5880	Merchant Transmission	0.5880
276166	Z1-106 BAT1	0.5901	Merchant Transmission	0.5901
290021	O50 E	49.5168	50/50	49.5168
293771	O-035 E	2.7918	Adder	3.28
904211	W3-135	0.3981	50/50	0.3981
916211	Z1-072 E	2.1127	Adder	2.49
926821	AC1-168 C O1	0.5072	Adder	0.6
926822	AC1-168 E O1	3.4037	Adder	4.0
926841	AC1-171 C O1	0.4638	Adder	0.55
926842	AC1-171 E O1	3.0983	Adder	3.65
927201	AC1-214 C O1	0.8960	Adder	1.05
927202	AC1-214 E O1	2.8483	Adder	3.35
930751	AB1-122 2O1	320.2635	50/50	320.2635
934111	AD1-039 2	31.3858	50/50	31.3858
936291	AD2-038 C O1	1.2844	Adder	1.51
936292	AD2-038 E O1	8.5956	Adder	10.11
936511	AD2-066 C O1	16.0087	50/50	16.0087
936512	AD2-066 E O1	10.6725	50/50	10.6725
938851	AE1-113 C	20.4257	50/50	20.4257
938852	AE1-113 E	72.4183	50/50	72.4183
939321	AE1-163 C O1	3.2275	Adder	3.8
939322	AE1-163 E O1	19.8258	Adder	23.32
942421	AE2-255 C O1	7.7370	50/50	7.7370
942422	AE2-255 E O1	23.2110	50/50	23.2110
942651	AE2-281 C O1	0.4611	Adder	0.54
942652	AE2-281 E O1	2.8323	Adder	3.33
946541	AF1-318 C O1	3.3766	Adder	3.97
946542	AF1-318 E O1	15.8103	Adder	18.6
953201	J715 C	1.3555	PJM External (MISO)	1.3555
953202	J715 E	7.3335	PJM External (MISO)	7.3335
954981	J974 C	2.8337	PJM External (MISO)	2.8337
954982	J974 E	15.3313	PJM External (MISO)	15.3313
957751	AF2-069 C	0.1037	Adder	0.12
957752	AF2-069 E	0.3348	Adder	0.39
957761	AF2-070 C	0.1462	Adder	0.17
957762	AF2-070 E	0.7097	Adder	0.83
958341	AF2-128 C O1	0.9435	Adder	1.11

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
958342	AF2-128 E O1	4.4172	Adder	5.2
958491	AF2-143 C	5.9280	Adder	6.97
958492	AF2-143 E	3.9520	Adder	4.65
960382	AF2-329 BAT	2.9624	Merchant Transmission	2.9624
LGEE	LGEE	0.1244	Confirmed LTF	0.1244
CPL	CPL	0.1131	Confirmed LTF	0.1131
G-007A	G-007A	0.2158	Confirmed LTF	0.2158
VFT	VFT	0.5805	Confirmed LTF	0.5805
CBM-W2	CBM-W2	4.7584	Confirmed LTF	4.7584
TVA	TVA	0.4788	Confirmed LTF	0.4788
CBM-S2	CBM-S2	1.1271	Confirmed LTF	1.1271
CBM-S1	CBM-S1	2.7179	Confirmed LTF	2.7179

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
96265467	918050	AA1-018 TAP	CE	270769	GOODINGS ;2R	CE	1	COMED_P7_345-L1202__B-S+_345-L1227__R-S-A	tower	1494.0	97.34	101.66	AC	62.41

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274677	POWERTON ;5U	26.3031	50/50	26.3031
274678	POWERTON ;6U	26.3664	50/50	26.3664
274879	MINONK ;1U	1.3682	50/50	1.3682
290021	O50 E	32.5680	50/50	32.5680
293644	O22 E1	5.3373	Adder	6.28
293645	O22 E2	10.3605	Adder	12.19
294401	BSHIL;1U E	4.3316	Adder	5.1
294410	BSHIL;2U E	4.3316	Adder	5.1
918051	AA1-018 C OP	2.0581	50/50	2.0581
918052	AA1-018 E OP	81.9644	50/50	81.9644
925581	AC1-033 C	0.7092	Adder	0.83
925582	AC1-033 E	4.7478	Adder	5.59
926841	AC1-171 C O1	0.8747	Adder	1.03
926842	AC1-171 E O1	5.8427	Adder	6.87
930741	AB1-122 1O1	97.9225	50/50	97.9225
930751	AB1-122 2O1	117.0470	50/50	117.0470
934051	AD1-031 C O1	1.4394	Adder	1.69
934052	AD1-031 E O1	2.3484	Adder	2.76
934101	AD1-039 1	9.5964	50/50	9.5964
934111	AD1-039 2	11.4706	50/50	11.4706
934871	AD1-116 C	4.7734	50/50	4.7734
934872	AD1-116 E	7.7882	50/50	7.7882
936291	AD2-038 C O1	8.1132	50/50	8.1132
936292	AD2-038 E O1	54.2958	50/50	54.2958
938851	AE1-113 C	13.4343	50/50	13.4343
938852	AE1-113 E	47.6307	50/50	47.6307
939321	AE1-163 C O1	20.3869	50/50	20.3869
939322	AE1-163 E O1	125.2341	50/50	125.2341
942421	AE2-255 C O1	5.0887	50/50	5.0887
942422	AE2-255 E O1	15.2662	50/50	15.2662
942651	AE2-281 C O1	2.9124	50/50	2.9124
942652	AE2-281 E O1	17.8906	50/50	17.8906
951631	J456 C	1.4611	PJM External (MISO)	1.4611
951632	J456 E	7.9049	PJM External (MISO)	7.9049
954702	J844 E	9.7505	PJM External (MISO)	9.7505
954981	J974 C	5.1940	PJM External (MISO)	5.1940
954982	J974 E	28.1010	PJM External (MISO)	28.1010
955711	J1055 C	1.5673	PJM External (MISO)	1.5673
955712	J1055 E	8.4796	PJM External (MISO)	8.4796
958481	AF2-142 C	56.5272	50/50	56.5272
958482	AF2-142 E	37.6848	50/50	37.6848

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
958491	AF2-143 C	37.4454	50/50	37.4454
958492	AF2-143 E	24.9636	50/50	24.9636
959351	AF2-226 C	1.3081	Adder	1.54
959352	AF2-226 E	1.9622	Adder	2.31
960281	AF2-319 C	1.3081	Adder	1.54
960282	AF2-319 E	1.9622	Adder	2.31
990901	L-005 E	7.2746	Adder	8.56
LGEE	LGEE	0.1004	Confirmed LTF	0.1004
CPL	CPL	0.1515	Confirmed LTF	0.1515
CBM-W2	CBM-W2	12.5798	Confirmed LTF	12.5798
NY	NY	0.0658	Confirmed LTF	0.0658
TVA	TVA	1.3034	Confirmed LTF	1.3034
O-066	O-066	0.7123	Confirmed LTF	0.7123
CBM-S2	CBM-S2	1.9016	Confirmed LTF	1.9016
CBM-S1	CBM-S1	6.7308	Confirmed LTF	6.7308
G-007	G-007	0.1092	Confirmed LTF	0.1092
MADISON	MADISON	4.8626	Confirmed LTF	4.8626
MEC	MEC	5.0610	Confirmed LTF	5.0610
CBM-W1	CBM-W1	9.9455	Confirmed LTF	9.9455

11.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
96264009	270770	GOODINGS ;4B	CE	270766	GOODINGS ;3B	CE	1	COMED_P4_116-45-L11614_	breaker	1802.0	118.53	119.21	AC	21.56

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274658	DRESDEN ;2U	120.9372	Adder	142.28
274659	DRESDEN ;3U	86.3964	Adder	101.64
274734	ELWOOD EC;8P	4.9640	50/50	4.9640
274736	ELWOOD EC;9P	4.9640	50/50	4.9640
274830	U3-021 1	3.5373	Adder	4.16
274831	U3-021 2	3.5373	Adder	4.16
274859	EASYR;U1 E	6.6586	Adder	7.83
274860	EASYR;U2 E	6.6586	Adder	7.83
274861	TOP CROP ;1U	0.9778	50/50	0.9778
274862	TOP CROP ;2U	1.8981	50/50	1.8981
276167	Z1-106 E2	0.6840	Adder	0.8
276168	Z1-106 E1	0.6842	Adder	0.8
276170	Z1-108 E	1.4219	Adder	1.67
290021	O50 E	20.6271	Adder	24.27
293516	O-009 E1	5.7321	Adder	6.74
293517	O-009 E2	2.9115	Adder	3.43
293518	O-009 E3	3.2063	Adder	3.77
293644	O22 E1	23.2756	50/50	23.2756
293645	O22 E2	45.1820	50/50	45.1820
293715	O-029 E	6.1283	Adder	7.21
293716	O-029 E	3.3600	Adder	3.95
293717	O-029 E	3.0883	Adder	3.63
293771	O-035 E	4.1277	Adder	4.86
294401	BSHIL;1U E	5.7902	Adder	6.81
294410	BSHIL;2U E	5.7902	Adder	6.81
294763	P-046 E	5.5461	Adder	6.52
915011	Y3-013 1	2.2004	Adder	2.59
915021	Y3-013 2	2.2004	Adder	2.59
915031	Y3-013 3	2.2004	Adder	2.59
916211	Z1-072 E	3.1237	Adder	3.67
918052	AA1-018 E OP	15.6426	Adder	18.4
919581	AA2-030	10.8655	Adder	12.78
920272	AA2-123 E	1.4071	Adder	1.66
924471	AB2-096	24.5646	Adder	28.9
925581	AC1-033 C	0.9483	Adder	1.12
925582	AC1-033 E	6.3483	Adder	7.47
926311	AC1-109 1	1.0024	Adder	1.18
926321	AC1-109 2	1.0024	Adder	1.18
926331	AC1-110 1	1.0297	Adder	1.21
926341	AC1-110 2	1.0297	Adder	1.21
926431	AC1-114	1.3903	Adder	1.64
926841	AC1-171 C O1	0.8245	Adder	0.97

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
926842	AC1-171 E O1	5.5071	Adder	6.48
927091	AC1-204 1	18.4646	50/50	18.4646
927201	AC1-214 C O1	1.3248	Adder	1.56
927202	AC1-214 E O1	4.2114	Adder	4.95
927451	AC1-142A 1	3.3179	Adder	3.9
927461	AC1-142A 2	3.3391	Adder	3.93
927511	AC1-113 1	0.6951	Adder	0.82
927521	AC1-113 2	0.6951	Adder	0.82
930481	AB1-089	38.3490	Adder	45.12
930741	AB1-122 1O1	55.5758	Adder	65.38
930751	AB1-122 2O1	76.9537	Adder	90.53
932881	AC2-115 1	1.3903	Adder	1.64
932891	AC2-115 2	1.3903	Adder	1.64
932921	AC2-116	0.4866	Adder	0.57
933931	AD1-016 C	0.5347	Adder	0.63
933932	AD1-016 E	0.8724	Adder	1.03
934051	AD1-031 C O1	1.9250	Adder	2.26
934052	AD1-031 E O1	3.1408	Adder	3.7
934101	AD1-039 1	5.4464	Adder	6.41
934111	AD1-039 2	7.5415	Adder	8.87
934701	AD1-098 C O1	3.8027	Adder	4.47
934702	AD1-098 E O1	2.7763	Adder	3.27
934871	AD1-116 C	0.9110	Adder	1.07
934872	AD1-116 E	1.4864	Adder	1.75
934971	AD1-129 C	0.5241	Adder	0.62
934972	AD1-129 E	0.3494	Adder	0.41
935001	AD1-133 C O1	12.5582	Adder	14.77
935002	AD1-133 E O1	8.3722	Adder	9.85
936291	AD2-038 C O1	2.3823	Adder	2.8
936292	AD2-038 E O1	15.9433	Adder	18.76
936511	AD2-066 C O1	5.4125	Adder	6.37
936512	AD2-066 E O1	3.6084	Adder	4.25
936791	AD2-102 C	8.3283	Adder	9.8
936792	AD2-102 E	5.5522	Adder	6.53
937001	AD2-134 C	0.5418	Adder	0.64
937002	AD2-134 E	2.2382	Adder	2.63
937311	AD2-172 C	1.4558	Adder	1.71
937312	AD2-172 E	2.0105	Adder	2.37
937401	AD2-194 1	10.0439	Adder	11.82
937411	AD2-194 2	9.9904	Adder	11.75
937531	AD2-214 C	2.7938	Adder	3.29
937532	AD2-214 E	1.8625	Adder	2.19
938511	AE1-070 1	11.8016	Adder	13.88
938521	AE1-070 2	10.7397	Adder	12.63
938851	AE1-113 C	8.5087	Adder	10.01
938852	AE1-113 E	30.1672	Adder	35.49
938861	AE1-114 C O1	2.3513	Adder	2.77
938862	AE1-114 E O1	8.0221	Adder	9.44
939051	AE1-134 1	0.8443	Adder	0.99
939061	AE1-134 2	0.8443	Adder	0.99
939321	AE1-163 C O1	5.9864	Adder	7.04
939322	AE1-163 E O1	36.7733	Adder	43.26

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939683	AE1-198 BAT	26.8299	50/50	26.8299
940501	AE2-035 C	1.4558	Adder	1.71
940502	AE2-035 E	2.0105	Adder	2.37
942421	AE2-255 C O1	3.2230	Adder	3.79
942422	AE2-255 E O1	9.6690	Adder	11.38
942651	AE2-281 C O1	0.8552	Adder	1.01
942652	AE2-281 E O1	5.2533	Adder	6.18
942991	AE2-321 C	4.7098	Adder	5.54
942992	AE2-321 E	2.3197	Adder	2.73
943411	AF1-012 C	7.3185	Adder	8.61
943412	AF1-012 E	4.8790	Adder	5.74
943801	AF1-048 C	2.2016	Adder	2.59
943802	AF1-048 E	1.4678	Adder	1.73
943921	AF1-060	0.7168	Adder	0.84
944041	AF1-072	1.2136	Adder	1.43
946151	AF1-280 C O1	9.4965	Adder	11.17
946152	AF1-280 E O1	4.3670	Adder	5.14
946161	AF1-281 C	0.2080	Adder	0.24
946162	AF1-281 E	1.1784	Adder	1.39
946321	AF1-296 C O1	2.5356	Adder	2.98
946322	AF1-296 E O1	11.8710	Adder	13.97
946501	AF1-314 C	2.5867	Adder	3.04
946502	AF1-314 E	12.1104	Adder	14.25
946541	AF1-318 C O1	3.4852	Adder	4.1
946542	AF1-318 E O1	16.3187	Adder	19.2
946661	AF1-330 C	1.1538	Adder	1.36
946662	AF1-330 E	0.2533	Adder	0.3
957331	AF2-027 C	1.4669	Adder	1.73
957332	AF2-027 E	2.2004	Adder	2.59
957401	AF2-034 C	1.5319	Adder	1.8
957402	AF2-034 E	1.3049	Adder	1.54
957471	AF2-041 C	12.7158	Adder	14.96
957472	AF2-041 E	8.4772	Adder	9.97
957751	AF2-069 C	0.1534	Adder	0.18
957752	AF2-069 E	0.4951	Adder	0.58
957761	AF2-070 C	0.2161	Adder	0.25
957762	AF2-070 E	1.0494	Adder	1.23
958023	AF2-096 BAT	6.8551	Merchant Transmission	6.8551
958481	AF2-142 C	10.7880	Adder	12.69
958482	AF2-142 E	7.1920	Adder	8.46
958491	AF2-143 C	10.9953	Adder	12.94
958492	AF2-143 E	7.3302	Adder	8.62
958911	AF2-182 C	12.4772	Adder	14.68
958912	AF2-182 E	8.3181	Adder	9.79
958921	AF2-183 C	2.2182	Adder	2.61
958922	AF2-183 E	3.3272	Adder	3.91
959081	AF2-199 C	4.2386	Adder	4.99
959082	AF2-199 E	2.8257	Adder	3.32
959091	AF2-200 C O1	8.4772	Adder	9.97
959092	AF2-200 E O1	5.6515	Adder	6.65
959101	AF2-201 C O1	3.5634	Adder	4.19
959102	AF2-201 E O1	10.2325	Adder	12.04

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
959351	AF2-226 C	5.7048	50/50	5.7048
959352	AF2-226 E	8.5572	50/50	8.5572
960281	AF2-319 C	5.7048	50/50	5.7048
960282	AF2-319 E	8.5572	50/50	8.5572
960551	AF2-346 C	3.1365	Adder	3.69
960552	AF2-346 E	2.0910	Adder	2.46
960581	AF2-349 C	12.6332	Adder	14.86
960582	AF2-349 E	8.4221	Adder	9.91
960721	AF2-363 C O1	3.2501	Adder	3.82
960722	AF2-363 E O1	2.1668	Adder	2.55
960731	AF2-364 C O1	3.3983	Adder	4.0
960732	AF2-364 E O1	2.2656	Adder	2.67
961011	AF2-392 C O1	2.4295	Adder	2.86
961012	AF2-392 E O1	11.3745	Adder	13.38
961021	AF2-393 O1	4.1412	Adder	4.87
961031	AF2-394 O1	2.7608	Adder	3.25
990901	L-005 E	8.8791	Adder	10.45
BLUEG	BLUEG	0.7777	Confirmed LTF	0.7777
CBM-W2	CBM-W2	15.9377	Confirmed LTF	15.9377
NY	NY	0.5137	Confirmed LTF	0.5137
TVA	TVA	1.4042	Confirmed LTF	1.4042
WEC	WEC	2.6255	Confirmed LTF	2.6255
O-066	O-066	6.0077	Confirmed LTF	6.0077
CBM-S1	CBM-S1	6.3218	Confirmed LTF	6.3218
G-007	G-007	0.9277	Confirmed LTF	0.9277
MADISON	MADISON	13.7773	Confirmed LTF	13.7773
MEC	MEC	10.1632	Confirmed LTF	10.1632
TRIMBLE	TRIMBLE	0.2732	Confirmed LTF	0.2732
CATAWBA	CATAWBA	0.0364	Confirmed LTF	0.0364
CBM-W1	CBM-W1	39.2189	Confirmed LTF	39.2189

11.7.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
96265327	270855	POWERTON ;R	CE	349662	7TAZEWELL	AMIL	1	COMED_P7_345-L19601_B-S+_345-L9801__R-S_FSA	tower	1489.0	135.29	142.55	AC	110.8

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274677	POWERTON ;5U	95.2068	50/50	95.2068
274678	POWERTON ;6U	95.4420	50/50	95.4420
274861	TOP CROP ;1U	2.5260	50/50	2.5260
274862	TOP CROP ;2U	4.9035	50/50	4.9035
274879	MINONK ;1U	2.9974	50/50	2.9974
290021	O50 E	71.3504	50/50	71.3504
293644	O22 E1	60.1294	50/50	60.1294
293645	O22 E2	116.7218	50/50	116.7218
918051	AA1-018 C OP	2.4205	50/50	2.4205
918052	AA1-018 E OP	96.4003	50/50	96.4003
934871	AD1-116 C	5.6141	50/50	5.6141
934872	AD1-116 E	9.1599	50/50	9.1599
936291	AD2-038 C O1	14.4037	50/50	14.4037
936292	AD2-038 E O1	96.3938	50/50	96.3938
938851	AE1-113 C	29.4320	50/50	29.4320
938852	AE1-113 E	104.3500	50/50	104.3500
939321	AE1-163 C O1	36.1939	50/50	36.1939
939322	AE1-163 E O1	222.3337	50/50	222.3337
942421	AE2-255 C O1	11.1485	50/50	11.1485
942422	AE2-255 E O1	33.4455	50/50	33.4455
942651	AE2-281 C O1	5.1706	50/50	5.1706
942652	AE2-281 E O1	31.7620	50/50	31.7620
958481	AF2-142 C	66.4830	50/50	66.4830
958482	AF2-142 E	44.3220	50/50	44.3220
958491	AF2-143 C	66.4785	50/50	66.4785
958492	AF2-143 E	44.3190	50/50	44.3190
959351	AF2-226 C	14.7376	50/50	14.7376
959352	AF2-226 E	22.1064	50/50	22.1064
960281	AF2-319 C	14.7376	50/50	14.7376
960282	AF2-319 E	22.1064	50/50	22.1064
NEWTON	NEWTON	2.5337	Confirmed LTF	2.5337
BLUEG	BLUEG	1.8870	Confirmed LTF	1.8870
NY	NY	0.4396	Confirmed LTF	0.4396
PRAIRIE	PRAIRIE	8.1752	Confirmed LTF	8.1752
LGE-TSR-0092018	LGE-TSR-0092018	0.0000	LTF	0.0000
WEC	WEC	0.0684	Confirmed LTF	0.0684
O-066	O-066	5.3626	Confirmed LTF	5.3626
COFFEEN	COFFEEN	0.6564	Confirmed LTF	0.6564
CHEOAH	CHEOAH	0.7583	Confirmed LTF	0.7583
EDWARDS	EDWARDS	5.4516	Confirmed LTF	5.4516
TILTON	TILTON	1.1964	Confirmed LTF	1.1964

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
G-007	G-007	0.8310	Confirmed LTF	0.8310
GIBSON	GIBSON	0.7660	Confirmed LTF	0.7660
CALDERWOOD	CALDERWOOD	0.7569	Confirmed LTF	0.7569
FARMERCITY	FARMERCITY	0.2105	Confirmed LTF	0.2105
TRIMBLE	TRIMBLE	0.5899	Confirmed LTF	0.5899
CATAWBA	CATAWBA	0.4389	Confirmed LTF	0.4389

11.7.5 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
96265342	930740	AB1-122 TAP	CE	270717	DRESDEN ; R	CE	1	COMED_P7_345-L19601_B-S+_345-L9801_R-S_FSA	tower	1195.0	132.72	135.16	AC	32.08

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274677	POWERTON ;5U	27.5669	50/50	27.5669
274678	POWERTON ;6U	27.6457	50/50	27.6457
274861	TOP CROP ;1U	0.7308	50/50	0.7308
274862	TOP CROP ;2U	1.4186	50/50	1.4186
274879	MINONK ;1U	0.7783	50/50	0.7783
290021	O50 E	18.5264	50/50	18.5264
293644	O22 E1	17.3955	50/50	17.3955
293645	O22 E2	33.7677	50/50	33.7677
293771	O-035 E	3.9250	Adder	4.62
294401	BSHIL;1U E	5.8908	Adder	6.93
294410	BSHIL;2U E	5.8908	Adder	6.93
916211	Z1-072 E	2.9702	Adder	3.49
918051	AA1-018 C OP	0.7005	50/50	0.7005
918052	AA1-018 E OP	27.8996	50/50	27.8996
925581	AC1-033 C	0.9646	Adder	1.13
925582	AC1-033 E	6.4578	Adder	7.6
926821	AC1-168 C O1	0.5446	Adder	0.64
926822	AC1-168 E O1	3.6550	Adder	4.3
926841	AC1-171 C O1	1.3923	50/50	1.3923
926842	AC1-171 E O1	9.2997	50/50	9.2997
927201	AC1-214 C O1	1.2597	Adder	1.48
927202	AC1-214 E O1	4.0045	Adder	4.71
930741	AB1-122 1O1	524.4230	50/50	524.4230
934051	AD1-031 C O1	1.9578	Adder	2.3
934052	AD1-031 E O1	3.1943	Adder	3.76
934101	AD1-039 1	51.3935	50/50	51.3935
934871	AD1-116 C	1.6248	50/50	1.6248
934872	AD1-116 E	2.6510	50/50	2.6510
936291	AD2-038 C O1	4.1705	50/50	4.1705
936292	AD2-038 E O1	27.9100	50/50	27.9100
938851	AE1-113 C	7.6421	50/50	7.6421
938852	AE1-113 E	27.0949	50/50	27.0949
939321	AE1-163 C O1	10.4796	50/50	10.4796
939322	AE1-163 E O1	64.3749	50/50	64.3749
942421	AE2-255 C O1	2.8948	50/50	2.8948
942422	AE2-255 E O1	8.6843	50/50	8.6843
942651	AE2-281 C O1	1.4971	50/50	1.4971
942652	AE2-281 E O1	9.1964	50/50	9.1964
946541	AF1-318 C O1	2.9620	Adder	3.48
946542	AF1-318 E O1	13.8689	Adder	16.32
951631	J456 C	1.9539	PJM External (MISO)	1.9539

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
951632	J456 E	10.5711	PJM External (MISO)	10.5711
953951	J859	10.7522	PJM External (MISO)	10.7522
954702	J844 E	13.2756	PJM External (MISO)	13.2756
954981	J974 C	7.1109	PJM External (MISO)	7.1109
954982	J974 E	38.4716	PJM External (MISO)	38.4716
955711	J1055 C	2.0719	PJM External (MISO)	2.0719
955712	J1055 E	11.2093	PJM External (MISO)	11.2093
957751	AF2-069 C	0.1459	Adder	0.17
957752	AF2-069 E	0.4707	Adder	0.55
957761	AF2-070 C	0.2055	Adder	0.24
957762	AF2-070 E	0.9978	Adder	1.17
958341	AF2-128 C O1	0.7556	Adder	0.89
958342	AF2-128 E O1	3.5375	Adder	4.16
958481	AF2-142 C	19.2411	50/50	19.2411
958482	AF2-142 E	12.8274	50/50	12.8274
958491	AF2-143 C	19.2483	50/50	19.2483
958492	AF2-143 E	12.8322	50/50	12.8322
959351	AF2-226 C	4.2636	50/50	4.2636
959352	AF2-226 E	6.3954	50/50	6.3954
960281	AF2-319 C	4.2636	50/50	4.2636
960282	AF2-319 E	6.3954	50/50	6.3954
990901	L-005 E	9.8787	Adder	11.62
LGEE	LGEE	0.1301	Confirmed LTF	0.1301
CPL	CPL	0.2137	Confirmed LTF	0.2137
CBM-W2	CBM-W2	16.1589	Confirmed LTF	16.1589
NY	NY	0.0630	Confirmed LTF	0.0630
TVA	TVA	1.6954	Confirmed LTF	1.6954
O-066	O-066	0.6586	Confirmed LTF	0.6586
CBM-S2	CBM-S2	2.6010	Confirmed LTF	2.6010
CBM-S1	CBM-S1	8.7586	Confirmed LTF	8.7586
G-007	G-007	0.1009	Confirmed LTF	0.1009
MADISON	MADISON	6.4411	Confirmed LTF	6.4411
MEC	MEC	6.8263	Confirmed LTF	6.8263
CBM-W1	CBM-W1	14.6742	Confirmed LTF	14.6742

11.7.6 Index 6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
96265405	930750	AB1-122 TAP	CE	270716	DRESDEN ; B	CE	1	COMED_P7_345-L19601_B-S_+_345-L9801_R-S_FSA	tower	1768.0	106.08	108.01	AC	37.91

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274677	POWERTON ;5U	32.8986	50/50	32.8986
274678	POWERTON ;6U	32.6634	50/50	32.6634
274861	TOP CROP ;1U	0.8724	50/50	0.8724
274862	TOP CROP ;2U	1.6934	50/50	1.6934
274879	MINONK ;1U	3.6661	50/50	3.6661
290021	O50 E	87.2672	50/50	87.2672
293644	O22 E1	20.7656	50/50	20.7656
293645	O22 E2	40.3096	50/50	40.3096
918051	AA1-018 C OP	0.8279	50/50	0.8279
918052	AA1-018 E OP	32.9721	50/50	32.9721
926841	AC1-171 C O1	1.1539	50/50	1.1539
926842	AC1-171 E O1	7.7077	50/50	7.7077
930751	AB1-122 2O1	520.5245	50/50	520.5245
934111	AD1-039 2	51.0114	50/50	51.0114
934871	AD1-116 C	1.9202	50/50	1.9202
934872	AD1-116 E	3.1330	50/50	3.1330
936291	AD2-038 C O1	4.9278	50/50	4.9278
936292	AD2-038 E O1	32.9787	50/50	32.9787
938851	AE1-113 C	35.9977	50/50	35.9977
938852	AE1-113 E	127.6283	50/50	127.6283
939321	AE1-163 C O1	12.3828	50/50	12.3828
939322	AE1-163 E O1	76.0657	50/50	76.0657
942421	AE2-255 C O1	13.6355	50/50	13.6355
942422	AE2-255 E O1	40.9065	50/50	40.9065
942651	AE2-281 C O1	1.7690	50/50	1.7690
942652	AE2-281 E O1	10.8665	50/50	10.8665
951631	J456 C	1.6771	PJM External (MISO)	1.6771
951632	J456 E	9.0734	PJM External (MISO)	9.0734
954702	J844 E	10.8839	PJM External (MISO)	10.8839
954981	J974 C	5.9623	PJM External (MISO)	5.9623
954982	J974 E	32.2577	PJM External (MISO)	32.2577
955711	J1055 C	1.8133	PJM External (MISO)	1.8133
955712	J1055 E	9.8104	PJM External (MISO)	9.8104
958481	AF2-142 C	22.7394	50/50	22.7394
958482	AF2-142 E	15.1596	50/50	15.1596
958491	AF2-143 C	22.7439	50/50	22.7439
958492	AF2-143 E	15.1626	50/50	15.1626
959351	AF2-226 C	5.0896	50/50	5.0896
959352	AF2-226 E	7.6344	50/50	7.6344
960281	AF2-319 C	5.0896	50/50	5.0896
960282	AF2-319 E	7.6344	50/50	7.6344

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
990901	L-005 E	7.8234	Adder	9.2
LGEE	LGEE	0.1421	Confirmed LTF	0.1421
CPL	CPL	0.1951	Confirmed LTF	0.1951
CBM-W2	CBM-W2	14.0540	Confirmed LTF	14.0540
NY	NY	0.0437	Confirmed LTF	0.0437
TVA	TVA	1.4700	Confirmed LTF	1.4700
O-066	O-066	0.4435	Confirmed LTF	0.4435
CBM-S2	CBM-S2	2.3293	Confirmed LTF	2.3293
CBM-S1	CBM-S1	7.6765	Confirmed LTF	7.6765
G-007	G-007	0.0676	Confirmed LTF	0.0676
MADISON	MADISON	4.7134	Confirmed LTF	4.7134
MEC	MEC	5.1801	Confirmed LTF	5.1801
CBM-W1	CBM-W1	7.9188	Confirmed LTF	7.9188

11.7.7 Index 7

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	ACID C	MW IMPACT
95294072	960680	AF2-359 TAP	CE	243229	05OLIVE	AEP	1	AEP_P4_#2978_05DUMONT_FSA	breaker	971.0	165.73	165.75	AC	15.37

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
274658	DRESDEN ;2U	80.9931	Adder	95.29
274659	DRESDEN ;3U	78.2741	Adder	92.09
274806	UNIV PK N;2U	1.7950	50/50	1.7950
274808	UNIV PK N;4U	1.7950	50/50	1.7950
274809	UNIV PK N;5U	1.7950	50/50	1.7950
274811	UNIV PK N;7U	1.7950	50/50	1.7950
274812	UNIV PK N;8U	1.7950	50/50	1.7950
274814	UNIV PK N;0U	1.7950	50/50	1.7950
274815	UNIV PK N;XU	1.7950	50/50	1.7950
274881	PILOT HIL;1E	12.8092	Adder	15.07
275149	KELLYCK ;1E	12.8092	Adder	15.07
276167	Z1-106 E2	0.8833	Adder	1.04
276168	Z1-106 E1	0.8832	Adder	1.04
276169	Z1-107 E	1.8746	Adder	2.21
276170	Z1-108 E	1.7399	Adder	2.05
290108	LEEDK;1U E	16.9762	Adder	19.97
293061	N-015 E	10.8783	Adder	12.8
294392	P-010 E	13.8154	Adder	16.25
915011	Y3-013 1	2.6237	Adder	3.09
915021	Y3-013 2	2.6237	Adder	3.09
915031	Y3-013 3	2.6237	Adder	3.09
918052	AA1-018 E OP	12.3460	Adder	14.52
920272	AA2-123 E	1.7073	Adder	2.01
924471	AB2-096	25.1401	Adder	29.58
926311	AC1-109 1	1.3348	Adder	1.57
926321	AC1-109 2	1.3348	Adder	1.57
926331	AC1-110 1	1.3274	Adder	1.56
926341	AC1-110 2	1.3274	Adder	1.56
926351	AC1-111 1	0.5339	Adder	0.63
926361	AC1-111 2	0.5339	Adder	0.63
926371	AC1-111 3	0.5339	Adder	0.63
926381	AC1-111 4	0.5339	Adder	0.63
926391	AC1-111 5	0.5339	Adder	0.63
926401	AC1-111 6	0.5339	Adder	0.63
927451	AC1-142A 1	2.9754	Adder	3.5
927461	AC1-142A 2	2.9743	Adder	3.5
930501	AB1-091 O1	49.2324	Adder	57.92
930741	AB1-122 1O1	50.0773	Adder	58.91
930751	AB1-122 2O1	51.4947	Adder	60.58
932931	AC2-117	8.9500	Adder	10.53
933411	AC2-154 C	1.7385	Adder	2.05

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
933412	AC2-154 E	2.8366	Adder	3.34
933431	AC2-156 C O1	0.6673	Adder	0.79
933432	AC2-156 E O1	1.0887	Adder	1.28
933931	AD1-016 C	0.6488	Adder	0.76
933932	AD1-016 E	1.0585	Adder	1.25
934101	AD1-039 1	4.9076	Adder	5.77
934111	AD1-039 2	5.0465	Adder	5.94
934721	AD1-100 C	13.4105	Adder	15.78
934722	AD1-100 E	62.5821	Adder	73.63
934871	AD1-116 C	0.7190	Adder	0.85
934872	AD1-116 E	1.1731	Adder	1.38
936291	AD2-038 C O1	1.6984	Adder	2.0
936292	AD2-038 E O1	11.3665	Adder	13.37
936371	AD2-047 C O1	3.1111	Adder	3.66
936372	AD2-047 E O1	15.1894	Adder	17.87
936461	AD2-060	1.8300	Adder	2.15
936511	AD2-066 C O1	5.9018	Adder	6.94
936512	AD2-066 E O1	3.9345	Adder	4.63
937401	AD2-194 1	5.4820	Adder	6.45
937411	AD2-194 2	5.4866	Adder	6.45
938511	AE1-070 1	6.4413	Adder	7.58
938521	AE1-070 2	5.8981	Adder	6.94
939321	AE1-163 C O1	4.2679	Adder	5.02
939322	AE1-163 E O1	26.2170	Adder	30.84
939351	AE1-166 C O1	7.1306	Adder	8.39
939352	AE1-166 E O1	6.5821	Adder	7.74
940752	AE2-062 E	0.0945	Adder	0.11
941131	AE2-107 C	4.5836	Adder	5.39
941132	AE2-107 E	3.0557	Adder	3.59
941551	AE2-152 C O1	8.2276	Adder	9.68
941552	AE2-152 E O1	5.4851	Adder	6.45
941561	AE2-153 C O1	3.2966	Adder	3.88
941562	AE2-153 E O1	15.4340	Adder	18.16
942651	AE2-281 C O1	0.6097	Adder	0.72
942652	AE2-281 E O1	3.7453	Adder	4.41
942991	AE2-321 C	5.7109	Adder	6.72
942992	AE2-321 E	2.8129	Adder	3.31
943121	AE2-341 C	8.9093	Adder	10.48
943122	AE2-341 E	4.3749	Adder	5.15
943591	AF1-030 C O1	5.9248	Adder	6.97
943592	AF1-030 E O1	2.9314	Adder	3.45
943801	AF1-048 C	2.6697	Adder	3.14
943802	AF1-048 E	1.7798	Adder	2.09
944041	AF1-072	1.5059	Adder	1.77
944911	AF1-156 C	8.4288	Adder	9.92
944912	AF1-156 E	5.6192	Adder	6.61
945351	AF1-200 FTIR	215.5879	Merchant Transmission	215.5879
946661	AF1-330 C	1.4000	Adder	1.65
946662	AF1-330 E	0.3073	Adder	0.36
957331	AF2-027 C	1.7491	Adder	2.06
957332	AF2-027 E	2.6237	Adder	3.09
957401	AF2-034 C	0.9687	Adder	1.14

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
957402	AF2-034 E	0.8252	Adder	0.97
958011	AF2-095 C O1	12.4420	Adder	14.64
958012	AF2-095 E O1	5.8551	Adder	6.89
958021	AF2-096 C	23.9119	Adder	28.13
958022	AF2-096 E	11.2526	Adder	13.24
958481	AF2-142 C	8.5144	Adder	10.02
958482	AF2-142 E	5.6763	Adder	6.68
958491	AF2-143 C	7.8390	Adder	9.22
958492	AF2-143 E	5.2260	Adder	6.15
960381	AF2-329	4.6229	Adder	5.44
960581	AF2-349 C	15.2051	Adder	17.89
960582	AF2-349 E	10.1368	Adder	11.93
960591	AF2-350 C O1	5.4896	Adder	6.46
960592	AF2-350 E O1	3.6598	Adder	4.31
960601	AF2-351 C O1	0.7320	Adder	0.86
960602	AF2-351 E O1	1.0979	Adder	1.29
960681	AF2-359 C	63.9922	50/50	63.9922
960682	AF2-359 E	42.6615	50/50	42.6615
960721	AF2-363 C O1	4.1257	Adder	4.85
960722	AF2-363 E O1	2.7504	Adder	3.24
960731	AF2-364 C O1	4.2946	Adder	5.05
960732	AF2-364 E O1	2.8631	Adder	3.37
960751	AF2-366 C	4.8631	Adder	5.72
960752	AF2-366 E	3.2421	Adder	3.81
BLUEG	BLUEG	1.1406	Confirmed LTF	1.1406
CBM-W2	CBM-W2	16.2408	Confirmed LTF	16.2408
NY	NY	0.8627	Confirmed LTF	0.8627
TVA	TVA	1.1774	Confirmed LTF	1.1774
WEC	WEC	3.0870	Confirmed LTF	3.0870
O-066	O-066	10.1741	Confirmed LTF	10.1741
CBM-S1	CBM-S1	4.8394	Confirmed LTF	4.8394
G-007	G-007	1.5725	Confirmed LTF	1.5725
MADISON	MADISON	15.6704	Confirmed LTF	15.6704
MEC	MEC	10.6129	Confirmed LTF	10.6129
TRIMBLE	TRIMBLE	0.3946	Confirmed LTF	0.3946
CATAWBA	CATAWBA	0.1799	Confirmed LTF	0.1799
CBM-W1	CBM-W1	47.4504	Confirmed LTF	47.4504

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA1-018	Powerton-Goodings Grove	In Service
AA2-030	Nelson	Active
AA2-123	Marengo 34kV	In Service
AB1-089	Byron-Wayne 345kV #1	Active
AB1-091	Davis Creek 345kV	Active
AB1-122	Kendall-Tazewell & Dresden-Mole Creek	Engineering and Procurement
AB2-096	Silver Lake-Cherry Valley	Active
AC1-033	Kewanee	Active
AC1-109	Aurora 345kV	In Service
AC1-110	Aurora 138kV	Active
AC1-111	Aurora 138kV	In Service
AC1-113	Rockford	Partially in Service - Under Construction
AC1-114	Rockford II	Partially in Service - Under Construction
AC1-142A	Joliet	Active
AC1-168	Kewanee-Streator	Active
AC1-171	Powerton	Active
AC1-204	Elwood	Under Construction
AC1-214	Crescent Ridge	Partially in Service - Under Construction
AC2-115	Rockford	Partially in Service - Under Construction
AC2-116	Rockford II	Partially in Service - Under Construction
AC2-117	University Park North	Engineering and Procurement
AC2-154	Davis Creek 138kV	Active
AC2-156	Sandwich 34.5kV	Active
AD1-016	Marengo	Active
AD1-031	Kewanee 138 kV	Active
AD1-039	Kendall-Tazewell & Dresden-Mole Creek	Active
AD1-098	Dixon-McGirr	Active
AD1-100	Loretto-Wilton & Braidwood-Davis Creek	Active
AD1-116	Nevada 345 kV	Active
AD1-129	Belvidere 34 kV	Active
AD1-133	Pontiac MidPoint-Dresden	Active
AD2-038	Powerton	Active
AD2-047	Davis Creek 138 kV	Active
AD2-060	Davis Creek 138kV	Active
AD2-066	Mazon-Crescent Ridge	Active
AD2-102	Wempletown 345 kV	Active
AD2-134	Shady Oaks	Active
AD2-172	Lena 138kV	Active
AD2-194	Elwood	Active

Queue Number	Project Name	Status
AD2-214	Rock Falls-Garden Plains	Active
AE1-070	Elwood 345 kV	Active
AE1-113	Mole Creek 345 kV	Active
AE1-114	Maryland-Lancaster 138 kV	Active
AE1-134	Nelson 345 kV	Active
AE1-163	Powerton-Nevada 345 kV	Active
AE1-166	Loretto-Wilton & Braidwood-Davis Creek	Active
AE1-198	Crete 345 kV	Active
AE2-035	Lena 138 kV	Active
AE2-062	Romeoville 12 kV	Active
AE2-107	Haumesser Road 138 kV	Active
AE2-152	Loretto-Wilton & Braidwood-Davis Creek	Active
AE2-153	Braidwood-Davis Creek	Active
AE2-255	Molecreek 345 kV	Active
AE2-281	Powerton-Nevada 345 kV	Active
AE2-321	Belvidere-Marengo 138 kV	Active
AE2-341	Sandwich-Plano	Active
AF1-012	Electric Junc-Nelson	Active
AF1-030	Plano-R 138 kV	Active
AF1-048	Belvidere-Marengo	Active
AF1-060	Lena 138 kV	Active
AF1-072	Rocky Road	Active
AF1-156	Braidwood-Davis Creek	Active
AF1-200	Plano 345 kV	Active
AF1-280	Nelson-Lee County	Active
AF1-281	Nelson-Lee County	Active
AF1-296	Garden Plain 138 kV	Active
AF1-314	Lena 138 kV	Active
AF1-318	Crescent Ridge-Corbin	Active
AF1-330	Marengo	Active
AF2-027	Zion Energy Center 345 kV	Active
AF2-034	Kendall	Active
AF2-041	Nelson-Electric Junction 345 kV	Active
AF2-069	Crescent Ridge 138 kV	Active
AF2-070	Crescent Ridge 138 kV	Active
AF2-095	Wilmington-Davis Creek	Active
AF2-096	Braidwood-East Frankfort 345 kV	Active
AF2-128	Crescent Ridge-Corbin 138 kV	Active
AF2-142	Nevada 345 kV	Active
AF2-143	Powerton-Nevada 345 kV	Active
AF2-182	Nelson-Lee County 345 kV II	Active
AF2-183	Nelson-Lee County 345 kV	Active
AF2-199	Nelson-Electric Junction 345 kV	Active
AF2-200	Nelson-Electric Junction 345 kV	Active
AF2-201	Lena-Ecogrove 138 kV	Active
AF2-226	Katydid Road 345 kV	Active
AF2-319	Katydid Road 345 kV	Active
AF2-329	Sandwich-Plano 138 kV	Active
AF2-346	Electric Junction-Nelson 345 kV	Active
AF2-349	Garden Prairie-Silver Lake 345 kV	Active
AF2-350	Kensington 138 kV	Active
AF2-351	Kensington 138 kV	Active

Queue Number	Project Name	Status
AF2-359	Olive-University Park 345 kV	Active
AF2-363	Glidden 138 kV	Active
AF2-364	Wayne Red 138 kV	Active
AF2-366	Glidden-Waterman 138 kV	Active
AF2-392	Nelson-Dixon 138 kV	Active
AF2-393	Nelson-Dixon 138 kV	Active
AF2-394	Nelson-Dixon 138 kV	Active
U3-021	Silver Lake-Cherry Valley	Active
W3-135	Goose Lake 34.5kV	Under Construction
Y3-013	Zion Energy Center	Active
Z1-072	Crescent Ridge	In Service
Z1-106	West Chicago 34kV	In Service
Z1-107	Joliet 34kV	In Service
Z1-108	McHenry 34kV	In Service
J1055	MISO	MISO
J456	MISO	MISO
J715	MISO	MISO
J844	MISO	MISO
J859	MISO	MISO
J974	MISO	MISO

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
COMED_P4_116-45-L9801__FSA	CONTINGENCY 'COMED_P4_116-45-L9801__FSA' TRIP BRANCH FROM BUS 918050 TO BUS 270769 CKT 1 / AA1-018 GOODINGS ;2R 345 DISCONNECT BUS 270769 / GOODI;2R 345 END
COMED_P7_345-L19601_B-S+_345-L9801__R-S_FSA	CONTINGENCY 'COMED_P7_345-L19601_B-S+_345-L9801__R-S_FSA' TRIP BRANCH FROM BUS 270790 TO BUS 270770 CKT 1 / KATYD; B 345 GOODI;4B 345 TRIP BRANCH FROM BUS 918050 TO BUS 270769 CKT 1 / AA1-018 GOODINGS ;2R 345 END
COMED_P7_345-L0301_B-S+_345-L9801__R-S_FSA	CONTINGENCY 'COMED_P7_345-L0301_B-S+_345-L9801__R-S_FSA' TRIP BRANCH FROM BUS 270854 TO BUS 270790 CKT 1 / POWER; B 345 KATYDID TRIP BRANCH FROM BUS 918050 TO BUS 270769 CKT 1 / AA1-018 GOODINGS ;2R 345 END

Contingency Name	Contingency Definition
AEP_P4_#2978_05DUMONT_FSA	CONTINGENCY 'AEP_P4_#2978_05DUMONT_FSA' OPEN BRANCH FROM BUS 243206 TO BUS 907040 CKT 1 / 243206 05DUMONT 765 X1-020 OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTON ; 765 1 END
COMED_P7_345-L1202__B-S+_345-L1227__R-S-A	CONTINGENCY 'COMED_P7_345-L1202__B-S+_345-L1227__R-S-A' TRIP BRANCH FROM BUS 270716 TO BUS 930750 CKT 1 / DRESDEN ; B 345 AB1-122 TAP 345 TRIP BRANCH FROM BUS 930740 TO BUS 270717 CKT 1 / AB1-122 TAP 345 DRESDEN ; R 345 END
COMED_P4_012-45-BT4-5__	CONTINGENCY 'COMED_P4_012-45-BT4-5__' TRIP BRANCH FROM BUS 270716 TO BUS 930750 CKT 1 / DRESDEN ; B 345 AB1-122 TAP 345 TRIP BRANCH FROM BUS 270716 TO BUS 270736 CKT 1 / DRESD; B 345 ELWOO; B 345 TRIP BRANCH FROM BUS 270736 TO BUS 270737 CKT 1 / ELWOO; B 345 ELWOO; R 345 END
COMED_P4_116-45-L11614__	CONTINGENCY 'COMED_P4_116-45-L11614__' TRIP BRANCH FROM BUS 270667 TO BUS 270665 CKT 1 / B ISL;RT 345 B ISL; R 345 TRIP BRANCH FROM BUS 270667 TO BUS 270927 CKT 1 / B ISL;RT 345 WILTO; R 345 TRIP BRANCH FROM BUS 270769 TO BUS 270667 CKT 1 / GOODI;2R 345 B ISL;RT 345 DISCONNECT BUS 270769 / GOODI;2R 345 END
Base Case	
COMED_P4_012-45-BT5-6__	CONTINGENCY 'COMED_P4_012-45-BT5-6__' TRIP BRANCH FROM BUS 270716 TO BUS 270736 CKT 1 / DRESD; B 345 ELWOO; B 345 TRIP BRANCH FROM BUS 270736 TO BUS 270737 CKT 1 / ELWOO; B 345 ELWOO; R 345 TRIP BRANCH FROM BUS 274702 TO BUS 270716 CKT 1 / KEND
COMED_P2-2_116_GG-345R__2	CONTINGENCY 'COMED_P2-2_116_GG-345R__2' DISCONNECT BUS 270769 / GOODI;2R 345 END
COMED_P4_116-45-TR82__	CONTINGENCY 'COMED_P4_116-45-TR82__' TRIP BRANCH FROM BUS 270769 TO BUS 271565 TO BUS 275324 CKT 1 / GOODINGS ;2R 345 GOODINGS ; R 138 GOODINGS ;2C 34.5 DISCONNECT BUS 270769 / GOODINGS ;2R 345 END

Contingency Name	Contingency Definition
AEP_P4_#2978_05DUMONT_NON_FSA	CONTINGENCY 'AEP_P4_#2978_05DUMONT_NON_FSA' OPEN BRANCH FROM BUS 243206 TO BUS 907040 CKT 1 / 243206 05DUMONT 765 X1-020 OPEN BRANCH FROM BUS 243207 TO BUS 907040 CKT 1 OPEN BRANCH FROM BUS 243206 TO BUS 270644 CKT 1 / 243206 05DUMONT 765 270644 WILTON ; 765 1 END
COMED_P1-2_345-L1202__B-S-A	CONTINGENCY 'COMED_P1-2_345-L1202__B-S-A' TRIP BRANCH FROM BUS 270716 TO BUS 930750 CKT 1 / DRESDEN ; B 345 AB1-122 TAP 345 END
COMED_P1-2_345-L0304__R-S	CONTINGENCY 'COMED_P1-2_345-L0304__R-S' TRIP BRANCH FROM BUS 270855 TO BUS 349662 CKT 1 / POWER; R 345 7TAZEWELL 345 END

12 Short Circuit Analysis

No breakers were identified as overdutied as part of this study.

13 Stability and Reactive Power

Analysis to be run in the Facilities Study Phase.

14 Affected Systems

14.1 MISO

MISO Impacts to be determined during later study phases (as applicable).

Attachment 1: One Line Diagram

